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Thermodynamic assessments of the In–Eu and In–Yb systems

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ABSTRACT

Based on the experimental data including thermodynamic properties and phase equilibria, thermodynamic assessments of the In–Eu and In–Yb binary systems have been carried out by using the CALPHAD (Calculation of Phase Diagrams) method. The Gibbs free energies of the solution phases (liquid, fcc, bcc and tetragonal_A6) were described by subregular solution model with the Redlich–Kister equation. The intermetallic compounds, except the InYb phase, were treated as stoichiometric phases, and the InYb phase was described using the sublattice model. A consistent set of thermodynamic parameters has been derived for describing the Gibbs free energy of each solution phase and intermetallic compound in the In–Eu and In–Yb systems. A good agreement between the calculated results and the experimental data in the In–Eu and In–Yb systems is obtained.

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1. Introduction

The alloys of rare earths (Res) with elements of group III are very interesting as materials with applications as superconductors, catalysts and reaction promoting additives, and cathodic materials with various emission properties [1]. The phase diagrams, crystal structures, and lattice parameters of the In–Re systems were systematically studied in previous literature [1–7]. However, the In–Eu phase diagram, as well as the In–Yb phase diagram, is not similar to other In–Re phase diagrams, because Eu and Yb are divalent whereas other Re elements are trivalent [8].

In the present work, the thermodynamic assessments of the In–Eu and In–Yb systems have been carried out by means of the CALPHAD method based on the experimental information.

2. Experimental information

2.1. The In–Eu system

The accepted phase diagram of the In–Eu system was reviewed by Okamoto [8] on the basis of the experimental information [9–11]. In the In–Eu system, there are two terminal solution phases ((In) and (Eu)), and five intermetallic compounds (αInEu$_2$, βInEu$_2$, InEu, In$_2$Eu, and In$_4$Eu). The partial phase diagram of the In–Eu binary system in the composition range from 50 to 100 at.% In proposed earlier by Koster and Meixner [9], who determined three invariant reactions: L $\leftrightarrow$ InEu + In$_2$Eu at 860 °C, L $\leftrightarrow$ (In) + In$_4$Eu at 156 °C, and L + In$_2$Eu $\leftrightarrow$ In$_4$Eu at 445 °C. Yatsenko et al. [11] investigated the phase diagram of this system using thermal, metallographic, and X-ray analyses. In Yatsenko's work [11], the temperature of a peritectic reaction, L + In$_2$Eu $\leftrightarrow$ In$_4$Eu, was 450 °C, which was used in the present assessment considering the dependability. Yatsenko et al. [11] also found another eutectic reaction in this system, L $\leftrightarrow$ (Eu) + In$_2$Eu, and determined the eutectic point was at about 15 at.% In and 620 °C. Thermal analysis data indicated the existence of an allotropic transformation in InEu$_2$, αInEu$_2$ $\leftrightarrow$ βInEu$_2$ at 660 °C, and the peritectic melting point of βInEu$_2$ was 710 °C [11].

In this binary system there are two intermetallic compounds, InEu and In$_2$Eu, which have congruent melting temperatures. The compound In$_2$Eu was first found by Iandelli [3]. The congruent melting temperature of the In$_2$Eu phase is 980 °C determined by Koster et al. [9] and Yatsenko et al. [11]. The congruent melting temperature of InEu was reported to be 908 °C by Koster et al. [9] and 920 °C by Yatsenko et al. [11]. The later results were used in the present assessment. The crystal structures and lattice parameter of the intermetallic compounds in this system are not reported except In$_2$Eu, whose space group is P6$_3$/mmc [3].

The solubility of Eu in liquid In in the temperature range from 400 to 500 °C is measured by Dieva [10]. The liquid composition at the melting point of In is 99.6 at.% In, which indicates that the solubility of Eu in the terminal solution (In) must be less than 0.4 at.% [8]. Therefore, the solubility of Eu in (In) was ignored

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Table 1
The invariant reactions in the In–Eu system including calculated and experimental results.

Reaction	Composition (at.% In)			Temperature (°C)	Reference
L ↔ (Eu)+αInEu$_2$	16.1	0	33.33	620	This work
	15	0	33.3	620	[11]
L+InEu ↔ βInEu$_2$	33.3	50	33.33	712	This work
	~33.3	50	33.3	710	[8]
L ↔ InEu + In$_2$Eu	58.2	50	66.7	863	This work
	53.5	50	66.7	860	[11]
L + In$_2$Eu ↔ In$_4$Eu	85.4	66.7	80	450	This work
	91	66.7	80	450	[11]
L ↔ (In) + In$_4$Eu	99.5	80	100	154	This work
	~100	80	100	156	[8]

in the assessment procedure. The solubility of In in the terminal solution (Eu) is not reported in the previous literature, and was also ignored in the review work by Okamoto [8]. Thus, we also ignored the solubility of In in the (Eu) in the present assessment. All the invariant reactions which appeared in the In–Eu binary system are listed in Table 1.

The thermodynamic properties of dilute In–Eu solutions in equilibrium with In$_2$Eu in the temperature range from 723 to 1023 K were determined by Dubinin et al. [12] using the emf method. The formation enthalpy of the In$_2$Eu phase is about 56.1 kJ/mol [12]. Bushmanov et al. [13] measured the mixing enthalpies of the liquid phase at 1300 K.

2.2. The In–Yb system

The phase diagram of the In–Yb system was systematically investigated by Mcmasters et al. [14] using X-ray diffraction, differential thermal, and metallographic methods. Five intermetallic compounds (In$_3$Yb, In$_2$Yb, InYb, InYb$_2$, and In$_2$Yb$_5$) were found [14], where In$_2$Yb$_5$, InYb$_2$, and In$_3$Yb decompose peritectically at 810 °C, 820°C, and 630 °C, respectively, whereas InYb and In$_2$Yb melt congruently at 1067 °C and 890 °C, respectively. Four intermetallic compounds In$_3$Yb, In$_2$Yb, In Yb$_2$, and In$_2$Yb$_5$ have no composition range, however the InYb shows a wide homogeneity range toward Yb-rich composition [14].

In this system, there are three eutectic reactions, L ↔ (βYb) + In$_2$Yb$_5$, L ↔ InYb+In$_2$Yb, and L ↔ In$_3$Yb+(In), occurring at 690 °C and 11.5 at.% In, 875 °C and 63.0 at.% In, and 154 °C and greater than 99 at.% In. Three peritectic reactions, L + InYb$_2$ ↔ In$_2$Yb$_5$, L + InYb ↔ InYb$_2$, and L + In$_2$Yb ↔ In$_3$Yb, occurring at 810 °C, 820 °C, and 630 °C were also determined by Mcmasters et al. [14]. From the experimental data [14], it is suggested that there is also an invariant reaction, ((γYb) ↔ (βYb) + L), occurring at 750 °C in the Yb-rich side.

The solubility of In (βYb) is about 1.0 at.% In at eutectic temperature (690 °C) and about 0.6 at.% In at 650 °C [14]. From the In-rich powder pattern data, the solubility of Yb in (In) is practically negligible [14]. The DTA results suggested that there is a polymorphic transformation of the InYb$_2$ phase between 670 °C (cooling arrests) and 695 °C (heating arrests), however, the metallographic results and powder pattern data showed no evidence of this transformation [14]. The reason may be that the transformation could be due to the effects of impurities and/or segregated melt [14]. Therefore, the possible polymorphic transformation was ignored in this assessment procedure.

Palenzona and Cirafici [15] measured the heat evolved in the formation of In$_3$Yb at 160 °C using a dynamic differential colorimetric method, and got the standard enthalpy of formation of the In$_3$Yb phase (39.7 kJ/mol) at 298 K.

Some research [3–5,14–18] reported the crystal structure and the lattice parameter data of the intermetallic compounds.

Table 2
The invariant reactions in the In–Yb system including calculated and experimental results.

Reaction	Composition (at.% In)			Temperature (°C)	References
(γYb) ↔ L + (βYb)	1	6	0.8	750	This work
	2	5	1.5	750	[14,19]
L ↔ (βYb)+In$_2$Yb$_5$	10.0	1.4	28.6	690	This work
	11.5	1	28.6	690	[14]
L+InYb$_2$ ↔ In$_2$Yb$_5$	28.3	33.3	28.6	810	This work
	25	33.3	28.6	810	[14]
L + InYb ↔ InYb$_2$	30.1	47.7	33.3	820	This work
	28	47	33.3	820	[14,19]
L ↔ InYb + In$_2$Yb	63.3	50	66.7	880	This work
	63	50	66.7	875	[14,19]
L + In$_2$Yb ↔ In$_3$Yb	83.3	66.7	75	630	This work
	86.5	66.7	75	630	[14,19]
L ↔ (In) + In$_3$Yb	99.97	100	75	156	This work
	>99	100	75	154	[14,19]

Palenzona and Cirafici [19] reviewed this system based on the previous literature. All the invariant reactions which appeared in the In–Yb binary system are listed in Table 2.

3. Thermodynamic models

3.1. Solution phases

The Gibbs free energies of the liquid, fcc (βYb), bcc ((γYb) and (Eu)) and tetragonal_A6 (In) phases in the In–Eu and In–Yb binary systems are described by the subregular solution model with the Redlich–Kister formula [20]. Taking the A–B system as the example, the Gibbs free energies of the solution phases are described by the following expression:

$$G_m^{\phi} = {}^0G_A^{\phi}x_A + {}^0G_B^{\phi}x_B + RT(x_A \ln x_A + x_B \ln x_B) + {}^EG^{\phi} \quad (1)$$

where G_i^{ϕ} is the molar Gibbs free energy of pure element i in the respective reference state with the ϕ phase, which is taken from the compilation by Dinsdale [21], and the term ${}^EG^{\phi}$ is the excess energy, which is expressed in the Redlich–Kister polynomial [20] as:

$$\begin{aligned} {}^EG^{\phi} &= {}^0L_{AB}^{\phi}x_Ax_B + {}^1L_{AB}^{\phi}(x_A - x_B)x_Ax_B + {}^2L_{AB}^{\phi}(x_A - x_B)^2x_Ax_B + \cdots \\ &= \sum_{m=0}^{n} {}^mL_{AB}^{\phi}(x_A - x_B)^m x_Ax_B \end{aligned} \quad (2)$$

where ${}^mL_{AB}^{\phi}$ is the interaction energy between A and B atoms, and expressed as:

$${}^mL_{AB}^{\phi} = a + bT \quad (3)$$

the parameters of a and b are evaluated based on the experimental data in the present work.

3.2. Intermetallic compounds

All the intermetallic compounds in the In–Eu and In–Yb systems, except the InYb phase, are treated as the stoichiometric compounds. The Gibbs free energy per mole of formula unit A_mB_n can be expressed as follows:

$$\Delta G_f^{A_mB_n} = G_f^{A_mB_n} - mG_A^{\mathrm{ref}} - nG_B^{\mathrm{ref}} = a' + b'T \quad (4)$$

where the $\Delta G_f^{A_mB_n}$ denotes the standard Gibbs free energy of formation of the stoichiometric compound from the pure

elements. The term G_i^{ref} is the molar Gibbs free energy of pure element i ($I = A, B$ respectively) with its defined reference structure. The parameters a' and b' are evaluated in the present work.

3.3. The InYb phase

As the InYb phase is an ordered phase B2_BCC in the In–Yb binary system, the InYb phase is described by the two-sublattice model ((In, Yb)$_{0.5}$(In, Yb)$_{0.5}$), and the Gibbs free energy can be expressed as follows:

$$G^{\mathrm{B2_BCC}} = \sum_{i,j=\mathrm{In,Yb}} Y_i^{\mathrm{I}} Y_j^{\mathrm{II}} G_{i:j}^{\mathrm{B2_BCC}} + 0.5RT\left[\sum_{i=\mathrm{In,Yb}} Y_i^{\mathrm{I}} \ln(Y_i^{\mathrm{I}}) + \sum_{i=\mathrm{In,Yb}} Y_i^{\mathrm{II}} \ln(Y_i^{\mathrm{II}})\right] + \sum_{i=\mathrm{In,Yb}} Y_{\mathrm{In}}^{\mathrm{I}} Y_{\mathrm{Yb}}^{\mathrm{I}} Y_i^{\mathrm{II}} L_{\mathrm{In,Yb}:i}^{\mathrm{B2_BCC}} + \sum_{j=\mathrm{In,Yb}} Y_{\mathrm{In}}^{\mathrm{II}} Y_{\mathrm{Yb}}^{\mathrm{II}} Y_j^{\mathrm{I}} L_{j:\mathrm{In,Yb}}^{\mathrm{B2_BCC}} \quad (5)$$

where $G_{i:j}^{\mathrm{B2_BCC}}$ ($i, j = \mathrm{In, Yb}$) represents the Gibbs free energies of the bcc (In), bcc (Yb) and of the hypothetical bcc compounds $\mathrm{In_{0.5}Yb_{0.5}}$ and $\mathrm{Yb_{0.5}In_{0.5}}$, respectively. y_i^{I} and y_i^{II} are the molar fractions of element i in the first and second sublattice, respectively. For the bcc structure, due to the symmetry of this model, it is assumed that $G_{\mathrm{In:Yb}}^{\mathrm{B2_BCC}} = G_{\mathrm{Yb:In}}^{\mathrm{B2_BCC}}$, and if the Neumann–Kopp rule is adopted, then

$$G_{\mathrm{In:Yb}}^{\mathrm{B2_BCC}} = a'' + b''T + 0.5G_{\mathrm{In}}^{\mathrm{ref}} + 0.5G_{\mathrm{Yb}}^{\mathrm{ref}}. \quad (6)$$

The interaction parameters $L_{\mathrm{In,Yb}:i}^{\mathrm{B2_BCC}}$ and $L_{j:\mathrm{In,Yb}}^{\mathrm{B2_BCC}}$ may take a similar form to those of solution phases. a'' and b'' are the thermodynamic parameters optimized in the present work.

4. Optimization results and discussion

Most of the above experimental information was selected for the evaluation of the thermodynamic model parameters. The optimization was carried out by means of the Thermo-Calc software developed by Sundman et al. [22]. The thermodynamic parameters of all phases in these two systems are obtained and listed in Table 3.

4.1. The In–Eu system

The calculated phase diagram of the In–Eu system together with the experimental data from [9–11] are shown in Fig. 1. From Fig. 1, it can be seen that the calculated phase diagram is in excellent agreement with the experimental data, except for the liquidus line in the In-rich side and the eutectic point of L ↔ InEu + $\mathrm{In_2Eu}$. The calculated liquidus line agrees well with the experimental data reported by Koster et al. [9], but it has somewhat of a deviation with the experimental data reported by Yatsenko et al. [11] and Dieva [10]. Considering the experimental error and the inconsistencies of the experimental data [9–11], the calculated liquidus line is acceptable. The calculated eutectic point of L ↔ InEu + $\mathrm{In_2Eu}$ is at 58.2 at.% In and 863 °C, which has a slight deviation with the results (53.5 at.% and 860 °C) reported by Yatsenko et al. [11]. The reason for this may be that the liquidus line experimentally determined in the Refs. [10,11] is visually asymmetrical near 50 at.% In, which is improbable thermodynamically [23]. Therefore, considering the symmetry of the liquidus line, the slight deviation between the calculated and experimental results is reasonable. The other eutectic points and peritectic points calculated in the present work in this system agree well with the experimental data, which are listed in Table 1. The calculated congruent melting temperatures of the InEu and $\mathrm{In_2Eu}$ phases are 920 °C and 980 °C, agreeing with the experimental data [9,11]. The allotropic transformation ($\alpha\mathrm{InEu_2} \leftrightarrow \beta\mathrm{InEu_2}$) at 660 °C reappears in the present work, as shown in Fig. 1. All the invariant reactions in this system together with the experimental data are listed in Table 1.

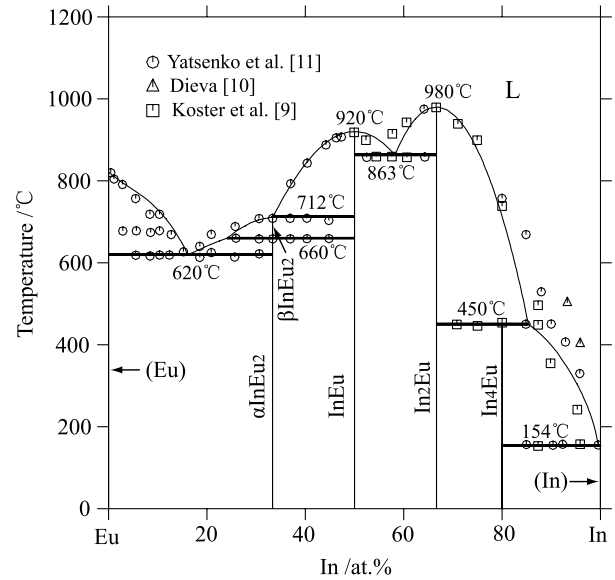


Fig. 1. The calculated phase diagram of the In–Eu system together with the experimental data [9–11].

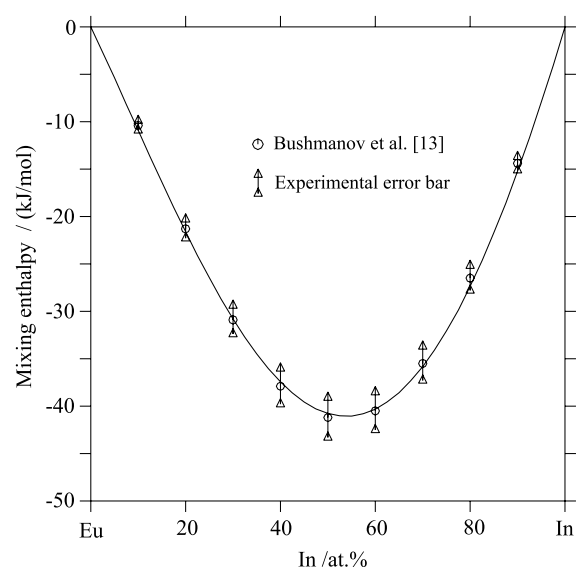


Fig. 2. The calculated mixing enthalpy of the liquid phase at 1300 K in the In–Eu system compared with the experimental data [13]. Reference states: Eu (liquid) and In (liquid).

The calculated mixing enthalpy of the liquid phase at 1300 K together with the experimental data determined by Bushmanov et al. [13] are shown in Fig. 2, from which it can be seen that a good agreement is obtained between the calculated result and experimental data. Fig. 3 shows the calculated formation enthalpies of the solid phases at 723 K compared with the experimental data [12]. It is seen that the calculated formation enthalpy of the $\mathrm{In_2Eu}$ phase is somewhat more positive than the experimental data, however, it is also acceptable.

Table 3
Thermodynamic parameters in the In–Eu and In–Yb systems optimized in the present work.

System	Phase	Thermodynamic parameters
In–Eu	Liquid phase	${}^{0}L^{\text{liquid}}_{\text{Eu,In}} = -163100 + 35.5T$
		${}^{1}L^{\text{liquid}}_{\text{Eu,In}} = 30000 - 15T$
		${}^{2}L^{\text{liquid}}_{\text{Eu,In}} = 28000$
	Bcc phase	${}^{0}L^{\text{bcc}}_{\text{Eu,In}} = 12000$
	Tetragonal_A6 phase	$G^{\text{Tetragonal_A6}}_{\text{Eu}} - G^{\text{ser}}_{\text{Eu}} = 1000$
	αInEu$_2$ phase	$G^{\alpha\text{InEu}_2}_{\text{In:Eu}} - G^{\text{ser}}_{\text{In}} - 2G^{\text{ser}}_{\text{Eu}} = -172804.5 + 76T$
	βInEu$_2$ phase	$G^{\beta\text{InEu}_2}_{\text{In:Eu}} - G^{\text{ser}}_{\text{In}} - 2G^{\text{ser}}_{\text{Eu}} = -172058 + 75.2T$
	InEu phase	$G^{\text{InEu}}_{\text{In:Eu}} - G^{\text{ser}}_{\text{In}} - G^{\text{ser}}_{\text{Eu}} = -117764 + 31T$
	In$_2$Eu phase	$G^{\text{In}_2\text{Eu}}_{\text{In:Eu}} - 2G^{\text{ser}}_{\text{In}} - G^{\text{ser}}_{\text{Eu}} = -137071 + 19T$
	In$_4$Eu phase	$G^{\text{In}_4\text{Eu}}_{\text{In:Eu}} - 4G^{\text{ser}}_{\text{In}} - G^{\text{ser}}_{\text{Eu}} = -188782 + 75.6T$
In–Yb	Liquid phase	${}^{0}L^{\text{liquid}}_{\text{In,Yb}} = -110000 + 2T$
		${}^{1}L^{\text{liquid}}_{\text{In,Yb}} = -2000 - 21T$
		${}^{2}L^{\text{liquid}}_{\text{In,Yb}} = 5000$
	Fcc phase	${}^{0}L^{\text{fcc}}_{\text{In.Yb}} = -65783$
	Bcc phase	${}^{0}L^{\text{bcc}}_{\text{In,Yb}} = -67500$
	Tetragonal_A6 phase	$G^{\text{Tetragonal_A6}}_{\text{Yb}} - G^{\text{ser}}_{\text{Yb}} = 1000$
	In$_3$Yb phase	$G^{\text{In}_3\text{Yb}}_{\text{In:Yb}} - 3G^{\text{ser}}_{\text{In}} - G^{\text{ser}}_{\text{Yb}} = -159550 + 44T$
	In$_2$Yb phase	$G^{\text{In}_2\text{Yb}}_{\text{In:Yb}} - 2G^{\text{ser}}_{\text{In}} - G^{\text{ser}}_{\text{Yb}} = -124910 + 15.1T$
	B2_BCC (InYb)	$G^{\text{B2_Bcc}}_{\text{In:In}} = G^{\text{B2_Bcc}}_{\text{Yb:Yb}} = 0$
	phase	$G^{\text{B2_Bcc}}_{\text{In:Yb}} = G^{\text{B2_Bcc}}_{\text{Yb:In}} = -42400 + 2.4T + 0.5G^{\text{ser}}_{\text{In}} + 0.5G^{\text{ser}}_{\text{Yb}}$
		${}^{0}L^{\text{B2_Bcc}}_{\text{In,Yb:Yb}} = {}^{0}L^{\text{B2_Bcc}}_{\text{Yb:In,Yb}} = -3000 - 76 * T$
		${}^{0}L^{\text{B2_Bcc}}_{\text{In,Yb:In}} = {}^{0}L^{\text{B2_Bcc}}_{\text{In:In,Yb}} = -30000$
	InYb$_2$ phase	$G^{\text{InYb}_2}_{\text{In:Yb}} - G^{\text{ser}}_{\text{In}} - 2G^{\text{ser}}_{\text{Yb}} = -124589 + 32.7T$
	In$_2$Yb$_5$ phase	$G^{\text{In}_2\text{Yb}_5}_{\text{In:Yb}} - 2G^{\text{ser}}_{\text{In}} - 5G^{\text{ser}}_{\text{Yb}} = -288280 + 95.1T$

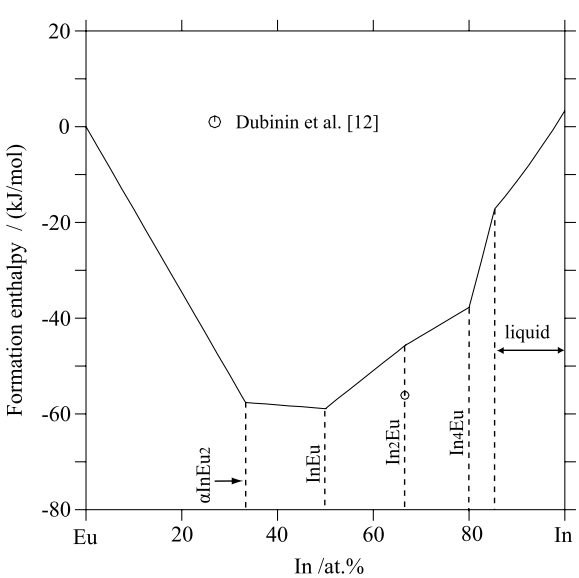


Fig. 3. The calculated formation enthalpies of the intermetallic phases at 723 K in the In–Eu system compared with the experimental data [12]. Reference states: Eu (bcc) and In (tetragonal_A6).

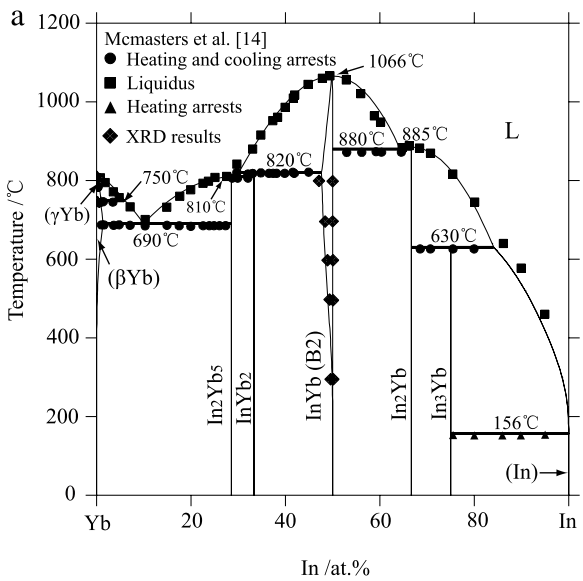
4.2. The In–Yb system

Fig. 4(a) is the calculated phase diagram of the In–Yb system together with the experimental data reported by Mcmasters et al. [14], and Fig. 4(b) is the magnified image of the phase diagram in the Yb-rich side. From Fig. 4(b), it can be seen that there is an invariant reaction ($(\gamma\text{Yb}) \leftrightarrow (\beta\text{Yb}) + L$) at 750 °C, which agrees well with the suggestion from Mcmasters et al. [14]. The calculated peritectic melting temperatures of the In$_2$Yb$_5$, InYb$_2$, and In$_3$Yb phases are 810 °C, 820 °C, and 630 °C, respectively, and the calculated congruent melting temperatures of the InYb and In$_2$Yb phases are 1066 °C and 885 °C, respectively. From comparison between the calculated results and experimental data, good agreement is obtained.

The calculated temperature of the eutectic reaction, L ↔ InYb+ In$_2$Yb, is 880 °C, which has a slight deviation with the experimental data (875 °C) [14]. Mcmasters et al. [14] suggested an experimental error (±5 °C) of the DTA results, therefore, the slight deviation (5 °C) is acceptable. The temperature of the eutectic reaction (L ↔ In$_3$Yb + (In)) calculated in the present work is 156 °C, which is 2 °C higher than the experimental value [14]. However, the deviation is within the experimental error (±5 °C) suggested by Mcmasters et al. [14].

The calculated solubility of In in (βYb) at eutectic temperature (690 °C) is 1.5 at.% In agreeing with the value reported by Mcmasters et al. [14]. The InYb phase is the only intermetallic compound, which has a composition range, in this system. The calculated composition range is shown in Fig. 4, from which we can see that the calculated result agrees well with the experimental data determined by Mcmasters et al. [14]. All the invariant reactions in this system together with the experimental data are listed in Table 2.

Fig. 5 shows the calculated formation enthalpies of the solid phases at 298 K together with the experimental data reported by Palenzona and Cirafici [15], where the reference states of the Yb and In are bcc and tetragonal_A6 structures. The calculated



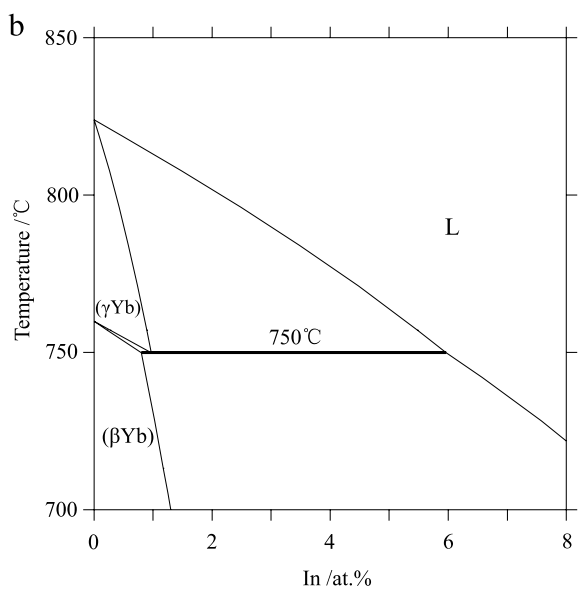


Fig. 4. (a) The calculated phase diagram of the In–Yb system together with the experimental data [14], and (b) the magnified image of the phase diagram in the Yb-rich side.

formation enthalpy of the In_3Yb phase is in excellent agreement with the experimental data [15].

5. Conclusion

The phase diagrams and thermodynamic properties of the In–Eu and In–Yb systems were thermodynamically assessed on the basis of the experimental data available in the literature. A consistent set of thermodynamic parameters for various thermodynamic models has been obtained, and the calculated peritectic and congruent melting temperatures of the intermetallic compounds in these two systems are in excellent agreement with the experimental data. The calculated phase equilibria and thermodynamic properties agree well with most of the literature data. From the present calculation, an invariant reaction ($(\gamma\text{Yb}) \leftrightarrow (\beta\text{Yb}) + L$) occurs at 750 °C in the Yb-rich side.

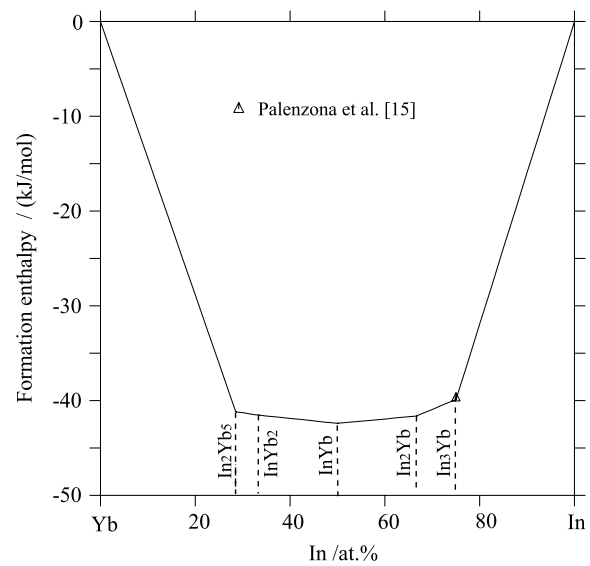


Fig. 5. The calculated formation enthalpies of the intermetallic phases at 298 K in the In–Yb system compared with the experimental data [15]. Reference states: Yb (fcc) and In (tetragonal_A6).

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Appendix. Supplementary data

Supplementary material related to this article can be found online at doi:10.1016/j.calphad.2010.10.003.

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